Homework Solution
Chapter 6: Annual Worth Analysis

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Exercise 1: Capital Recovery and Annual Worth of a Clinical Lab Analyzer

Solution:

Part 1. Using $(A/P, 9\%, 5) = 0.25709$ and $(A/F, 9\%, 5) = 0.16709$:

$$\begin{aligned} CR &= -380{,}000(A/P, 9\%, 5) + 40{,}000(A/F, 9\%, 5) \\ &= -\$91{,}011.44 \end{aligned}$$

Part 2.

$$\begin{aligned} AW &= CR + 115{,}000 \\ &= \$23{,}988.56 \end{aligned}$$

Part 3. Since $AW > 0$, the analyzer earns an equivalent extra profit of about \$23,988.56 every year **above and beyond** recovering its capital cost and the required 9% return. **The lab should proceed with the purchase.**

Exercise 2: Annual Worth of a Cost Alternative

Solution:

Using $(A/P, 7\%, 8) = 0.16747$:

$$\begin{aligned} CR &= -540{,}000(A/P, 7\%, 8) \\ &= -\$90{,}432.59 \end{aligned}$$

$$\begin{aligned} AW_{\text{cost}} &= CR - 65{,}000 \\ &= -\$155{,}432.59 \text{ per year} \end{aligned}$$

The Do-Nothing alternative is not realistic here because the de-icing service is **mandatory by regulation** — the airport must provide it regardless of cost, so there is no option to simply avoid the expenditure.

Exercise 3: Comparing Equal-Life Alternatives: AW and PW Together

Solution:

Part 1. Using $(P/A, 8\%, 6) = 4.62288$ and $(P/F, 8\%, 6) = 0.63017$:

$$\begin{aligned} PW_1 &= -720{,}000 + 60{,}000(P/F, 8\%, 6) + 185{,}000(P/A, 8\%, 6) \\ &= \$173{,}042.92 \end{aligned}$$

$$\begin{aligned} PW_2 &= -480{,}000 + 30{,}000(P/F, 8\%, 6) + 118{,}000(P/A, 8\%, 6) \\ &= \$84{,}404.89 \end{aligned}$$

Part 2. Using $(A/P, 8\%, 6) = 0.21632$:

$$\begin{aligned} AW_1 &= PW_1(A/P, 8\%, 6) \\ &= \$37{,}431.85 \end{aligned}$$

$$\begin{aligned} AW_2 &= PW_2(A/P, 8\%, 6) \\ &= \$18{,}258.08 \end{aligned}$$

Part 3. Using the direct Capital Recovery approach with $(A/F, 8\%, 6) = 0.13632$:

$$\begin{aligned} CR_1 &= -720{,}000(A/P, 8\%, 6) + 60{,}000(A/F, 8\%, 6) \\ &= -\$147{,}568.15 \end{aligned}$$

$$\begin{aligned} AW_1 &= CR_1 + 185{,}000 \\ &= \$37{,}431.85 \end{aligned}$$

$$\begin{aligned} CR_2 &= -480{,}000(A/P, 8\%, 6) + 30{,}000(A/F, 8\%, 6) \\ &= -\$99{,}741.92 \end{aligned}$$

$$\begin{aligned} AW_2 &= CR_2 + 118{,}000 \\ &= \$18{,}258.08 \end{aligned}$$

Both approaches agree exactly, confirming the mathematical equivalence of $PW \times (A/P, i, n)$ and the direct Capital Recovery method.

Part 4. Since $AW_1(\$37{,}431.85) > AW_2(\$18{,}258.08)$, **the university should select Option 1**.

Exercise 4: Independent Projects Screening via Annual Worth

Solution:

Part 1. Using $(A/P, 10\%, 6) = 0.22961$:

$$\begin{aligned} CR_1 &= -85{,}000(A/P, 10\%, 6) \\ &= -\$19{,}516.63 \end{aligned}$$

$$\begin{aligned} AW_1 &= CR_1 + 21{,}000 \\ &= \$1{,}483.37 \end{aligned}$$

$$\begin{aligned} CR_2 &= -140{,}000(A/P, 10\%, 6) \\ &= -\$32{,}145.03 \end{aligned}$$

$$\begin{aligned} AW_2 &= CR_2 + 29{,}000 \\ &= -\$3{,}145.03 \end{aligned}$$

$$\begin{aligned} CR_3 &= -45{,}000(A/P, 10\%, 6) \\ &= -\$10{,}332.33 \end{aligned}$$

$$\begin{aligned} AW_3 &= CR_3 + 11{,}500 \\ &= \$1{,}167.67 \end{aligned}$$

Part 2. Since $AW_1 > 0$ and $AW_3 > 0$, but $AW_2 < 0$, **the plant should implement Projects 1 and 3**, and reject Project 2.

Exercise 5: Mutually Exclusive Cost Alternatives with a Rising Operating

Cost

Solution:

Pump A. Using $(A/P, 6\%, 5) = 0.23740$:

$$\begin{aligned} CR_A &= -260{,}000(A/P, 6\%, 5) \\ &= -\$61{,}723.06 \end{aligned}$$

$$\begin{aligned} AW_A &= CR_A - 38{,}000 \\ &= -\$99{,}723.06 \end{aligned}$$

Pump B — Capital Recovery. Using $(A/P, 6\%, 8) = 0.16104$ and $(A/F, 6\%, 8) = 0.10104$:

$$\begin{aligned} CR_B &= -410{,}000(A/P, 6\%, 8) + 25{,}000(A/F, 6\%, 8) \\ &= -\$63{,}498.84 \end{aligned}$$

Pump B — Geometric operating cost. With $A_1 = \$15{,}000$, $g = 8\%$, $n = 8$, $i = 6\%$:

$$\begin{aligned} P &= A_1 \left[\frac{1 - (1+g)^n (1+i)^{-n}}{i - g} \right] \\ &= 15{,}000 \left[\frac{1 - (1.08)^8 (1.06)^{-8}}{0.06 - 0.08} \right] \\ &= \$120{,}972.38 \end{aligned}$$

$$\begin{aligned} AW_{\text{OpCost}} &= P(A/P, 6\%, 8) \\ &= \$19{,}480.90 \end{aligned}$$

Pump B — Total AW:

$$\begin{aligned} AW_B &= CR_B - AW_{\text{OpCost}} \\ &= -\$82{,}979.74 \end{aligned}$$

Decision. Since $AW_B(-\$82{,}979.74) > AW_A(-\$99{,}723.06)$, **Pump B should be selected**.

Exercise 6: Sensitivity of Capital Recovery to Salvage Value

Solution:

Using $(A/P, 8\%, 6) = 0.21632$ and $(A/F, 8\%, 6) = 0.13632$:

Part 1. Conservative salvage ($8,000):

$$\begin{aligned} CR_{\text{cons}} &= -65{,}000(A/P, 8\%, 6) + 8{,}000(A/F, 8\%, 6) \\ &= -\$12{,}969.98 \end{aligned}$$

Part 2. Optimistic salvage ($14,000):

$$\begin{aligned} CR_{\text{opt}} &= -65{,}000(A/P, 8\%, 6) + 14{,}000(A/F, 8\%, 6) \\ &= -\$12{,}152.08 \end{aligned}$$

Part 3. Percentage difference:

$$\frac{CR_{\text{opt}} - CR_{\text{cons}}}{CR_{\text{cons}}} \times 100\% = -6.31\%$$

A 75% increase in the salvage estimate ($8,000 → $14,000) changes Capital Recovery by only about 6.3%. In this case, CR is **relatively insensitive** to the salvage value estimate, since the salvage value is a small fraction of the $65,000 initial cost and is further discounted by $(A/F, 8\%, 6)$.

Exercise 7: Annual Worth of a Permanent Investment

Solution:

Part 1. Converting each component to an equivalent annual amount at $i = 5\%$:

Initial cost:

$$\begin{aligned} A_P &= -6{,}500{,}000 \times 0.05 \\ &= -\$325{,}000 \text{ per year} \end{aligned}$$

Routine annual upkeep:

$$A_{\text{maint}} = -\$85{,}000 \text{ per year}$$

Non-recurring foundation reinforcement at Year 8, using $(P/F, 5\%, 8) = 0.67684$:

$$\begin{aligned} A_C &= -400{,}000 \times \frac{0.05}{(1.05)^8} \\ &= -400{,}000(0.05)(0.67684) \\ &= -\$13{,}536.79 \text{ per year} \end{aligned}$$

Recurring resurfacing every 15 years, using $(A/F, 5\%, 15) = 0.04634$:

$$\begin{aligned} A_R &= -350{,}000(A/F, 5\%, 15) \\ &= -\$16{,}219.80 \text{ per year} \end{aligned}$$

Total:

$$\begin{aligned} AW_{\text{cost}} &= -325{,}000 - 85{,}000 - 13{,}536.79 - 16{,}219.80 \\ &= -\$439{,}756.59 \text{ per year} \end{aligned}$$

Part 2. This means that if the municipality sets aside approximately \$439,756.59 every single year, forever, that amount is exactly enough to cover the initial construction, routine upkeep, the one-time foundation reinforcement, and every future resurfacing cycle — the seawall can be built, maintained, and kept in service indefinitely without ever running short of funds.

Exercise 8: Relating Capitalized Cost to Annual Worth

Solution:

Using $AW = CC \times i$ (equivalently $CC = AW/i$) at $i = 6\%$:

Express Option 2's Annual Worth as a Capitalized Cost:

$$\begin{aligned} CC_2 &= \frac{135{,}000}{0.06} \\ &= \$2{,}250{,}000 \end{aligned}$$

Compare directly to Option 1's stated Capitalized Cost of \$2,400,000:

$$\begin{aligned} CC_1 &= \$2{,}400{,}000 \\ CC_2 &= \$2{,}250{,}000 \\ CC_1 &> CC_2 \end{aligned}$$

(Equivalently, expressing Option 1 as an Annual Worth: $AW_1 = 2{,}400{,}000 \times 0.06 = \$144{,}000 > AW_2 = \$135{,}000$.)

Option 1 requires the larger equivalent commitment from the university, whether compared as Capitalized Cost (\$2,400,000 vs. \$2,250,000) or as Annual Worth (\$144,000/yr vs. \$135,000/yr).

Exercise 9: Verifying Single-Life-Cycle AW Against LCM Present Worth

Solution:

Part 1 — Single-life-cycle AW. Using $(A/P, 9\%, 5) = 0.25709$, $(A/F, 9\%, 5) = 0.16709$, $(A/P, 9\%, 8) = 0.18067$, $(A/F, 9\%, 8) = 0.09067$:

$$\begin{aligned} CR_X &= -310{,}000(A/P, 9\%, 5) + 35{,}000(A/F, 9\%, 5) \\ &= -\$73{,}850.43 \end{aligned}$$

$$\begin{aligned} AW_X &= CR_X + 98{,}000 \\ &= \$24{,}149.57 \end{aligned}$$

$$\begin{aligned} CR_Y &= -450{,}000(A/P, 9\%, 8) + 50{,}000(A/F, 9\%, 8) \\ &= -\$76{,}769.75 \end{aligned}$$

$$\begin{aligned} AW_Y &= CR_Y + 112{,}000 \\ &= \$35{,}230.25 \end{aligned}$$

Part 2 — LCM Present Worth. LCM$(5, 8) = 40$ years. Truck X repeats 8 times (reinvestment at $t = 5, 10, 15, 20, 25, 30, 35$; salvage at $t = 5, 10, \ldots, 40$); Truck Y repeats 5 times (reinvestment at $t = 8, 16, 24, 32$; salvage at $t = 8, 16, 24, 32, 40$).

$$PW_X = -310{,}000\left[1 + \sum_{t=5,10,\ldots,35} (P/F, 9\%, t)\right] + 35{,}000 \sum_{t=5,10,\ldots,40} (P/F, 9\%, t) + 98{,}000(P/A, 9\%, 40)$$

$$PW_X = \$259{,}785.67$$

$$PW_Y = -450{,}000\left[1 + \sum_{t=8,16,24,32} (P/F, 9\%, t)\right] + 50{,}000 \sum_{t=8,16,24,32,40} (P/F, 9\%, t) + 112{,}000(P/A, 9\%, 40)$$

$$PW_Y = \$378{,}984.48$$

Part 3 — Converting LCM PW to AW. Using $(A/P, 9\%, 40) = 0.09296$:

$$\begin{aligned} AW_X &= PW_X(A/P, 9\%, 40) \\ &= \$24{,}149.57 \quad \checkmark \text{ (matches Part 1)} \end{aligned}$$

$$\begin{aligned} AW_Y &= PW_Y(A/P, 9\%, 40) \\ &= \$35{,}230.25 \quad \checkmark \text{ (matches Part 1)} \end{aligned}$$

Both methods agree exactly, confirming that a single life cycle is sufficient for AW analysis — there is no need to extend to the LCM.

Part 4. Since $AW_Y(\$35{,}230.25) > AW_X(\$24{,}149.57)$, **the company should purchase Truck Y**.

Exercise 10: Integrated Capstone: CapEx, OpEx, and a One-Time Cost

Solution:

Part 1. Using $(P/F, 8\%, 4) = 0.73503$ and $(A/P, 8\%, 7) = 0.19207$:

$$\begin{aligned} P_{\text{recal},0} &= -45{,}000(P/F, 8\%, 4) \\ &= -\$33{,}076.34 \end{aligned}$$

$$\begin{aligned} AW_{\text{recal}} &= P_{\text{recal},0}(A/P, 8\%, 7) \\ &= -\$6{,}353.05 \text{ per year} \end{aligned}$$

Part 2. Using $(A/F, 8\%, 7) = 0.11207$:

$$\begin{aligned} CR &= -920{,}000(A/P, 8\%, 7) + 60{,}000(A/F, 8\%, 7) \\ &= -\$169{,}982.27 \end{aligned}$$

No, the recalibration cost is **not** part of the CR. Capital Recovery only accounts for the initial investment (CapEx) and the salvage value — the Year 4 recalibration is a separate operating-type cost and is added afterward as its own annualized term.

Part 3. Combining CR, the annual net savings, and the annualized recalibration cost:

$$\begin{aligned} AW &= CR + 210{,}000 + AW_{\text{recal}} \\ &= -169{,}982.27 + 210{,}000 - 6{,}353.05 \\ &= \$33{,}664.68 \end{aligned}$$

Part 4. Using $(P/A, 8\%, 7) = 5.20637$:

$$\begin{aligned} PW &= AW(P/A, 8\%, 7) \\ &= \$175{,}270.79 \end{aligned}$$

Checking directly against the original cash flows (CapEx, salvage, annual savings, and the one-time Year 4 cost):

$$\begin{aligned} PW_{\text{direct}} &= -920{,}000 + 60{,}000(P/F, 8\%, 7) + 210{,}000(P/A, 8\%, 7) - 45{,}000(P/F, 8\%, 4) \\ &= \$175{,}270.79 \quad \checkmark \end{aligned}$$

Both methods agree exactly.

Part 5. Since $AW > 0$ (and equivalently $PW > 0$), **the hospital should proceed** with the sterile processing system — it exceeds the 8% MARR even after accounting for the mid-life recalibration cost.